

Fig. 6: Flow-chart of the program

1010 is sent for 54 ms and a signal of 1000 for the next 108 ms to transistors T1 through T4. Character 'D' is defined in the code to generate the 1010 and 1000 signals with 54 ms and 108 ms delays, respectively.

'A' left drift (Q1=1, Q2=0, Q3=1, Q4=0 for 54 ms) and Q1=0, Q2=0, Q3=1, Q4=0 for 108 ms). Left drift is possible by rotating the right motors at a high speed and the left motors at a low speed. This is possible with the PWM pulse given to the left motors. The left motors are given a pulse train of 33 per cent duty cycle so that these rotate at one-third the speed of the right motors. The car takes a left turn in forward direction, resulting in a drift. So signal 1010 is sent for 54 ms and signal 0010 for the next 108 ms to transistors T1 through T4. Character 'A' is defined in the code to generate 1010 and 0010 signals with 54ms and 108ms delays, respectively.

'Q' sharp-forward left turn (Q1=0, Q2=0, Q3=1, Q4=0). The toy car moves to the left sharply in the forward direction when the left motors are static and the right motors move clockwise. So signal 0010 is sent to transistors T1 through T4. Character 'Q' is defined

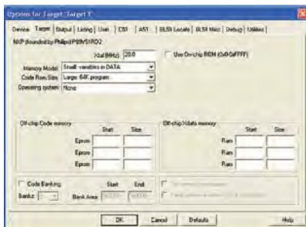


Fig. 7: Screenshot of 'Options for Target 1' window

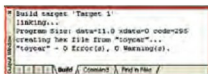


Fig. 8: Program compilation output screen

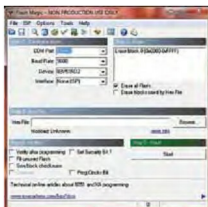


Fig. 9: Screenshot of 'Flash Magic' window

in the code to generate 0010 bits in the output.

'E' sharp-forward right turn (Q1=1, Q2=0, Q3=0, Q4=0). The car moves to the right sharply in the forward direction when the left motors move clockwise and the right motors are static. So signal 1000 is sent to transistors T1 through T4. Character 'E' is defined in the code to generate 1000 bits in the output.

'C' sharp-backward left turn (Q1=0, Q2=0, Q3=0, Q4=1). Sharp left turn takes place when the left motors are static and the right motors move anticlockwise. The car moves to the left

sharply in the backward direction. So signal 0001 is sent to transistors T1 through T4. Character 'C' is defined in the code to generate 0001 bits in the output.

'Z' sharp-backward right turn (Q1=0, Q2=1, Q3=0, Q4=0). The car moves to the right sharply in the backward direction when the left motors move anticlockwise and the right motors are static.

So signal 0100 is sent to transistors T1 through T4. Character 'Z' is defined in the code to generate 0100 bits in the output.

'U' stop (Q1=0, Q2=0, Q3=0, Q4=0). To stop the car, all the motors should be static. This is achieved by sending signal 0000 to the output of the microcontroller at its port pins p1.1 through p1.4. Character 'U' is defined in the code to generate 0000 bits in the output to stop the toy car.

Software program

Fig. 6 shows flow-chart of the program. The program is written in 'C' and compiled using Keil C software. The hex code generated using Keil software is burnt into the chip using Flash Magic programming software from NXP (Philips) Semiconductors.

Keil C uVision3 operations. 1. Run Keil uVision3 application from the desktop. From 'Project' menu, select 'New Project' option. Name the project as 'efytoy.uv2' and save it.

2. Select microcontroller P89V51RD2 from the database under NXP (Philips) option.

3. Right-click 'Source Group 1' option in 'Project Workspace' window on the left-hand side of the screen. Click 'Add Files to Source Group 1' option to add the toycar.c file.

4. Right-click 'Target 1' option from 'Project Workspace' and select 'Options for Target 1.' The window